



PROPAGATION ALGORITHMS FOR THE MINIMUM- DISTANCE CONSTRAINT

$\text{min_distance}(D, x, z, R)$

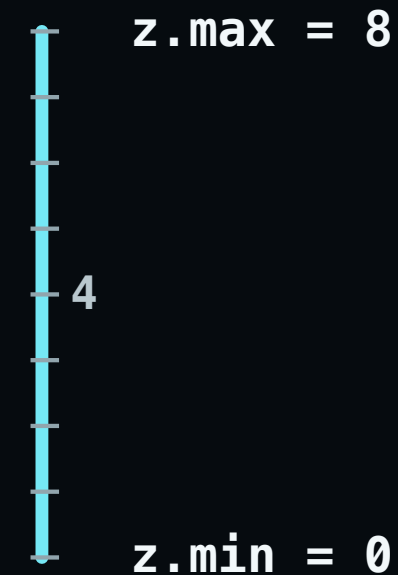
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CHOOSE FIVE SITES.



Euclidean distance

z range



Maximise the shortest distance among the five selected sites.

MODELLING MIN DISTANCE PROBLEMS

DIRECT DECOMPOSITION

$$y_{ij} = D[x_i, x_j] \quad z = \min_{i < j} (y_{ij})$$

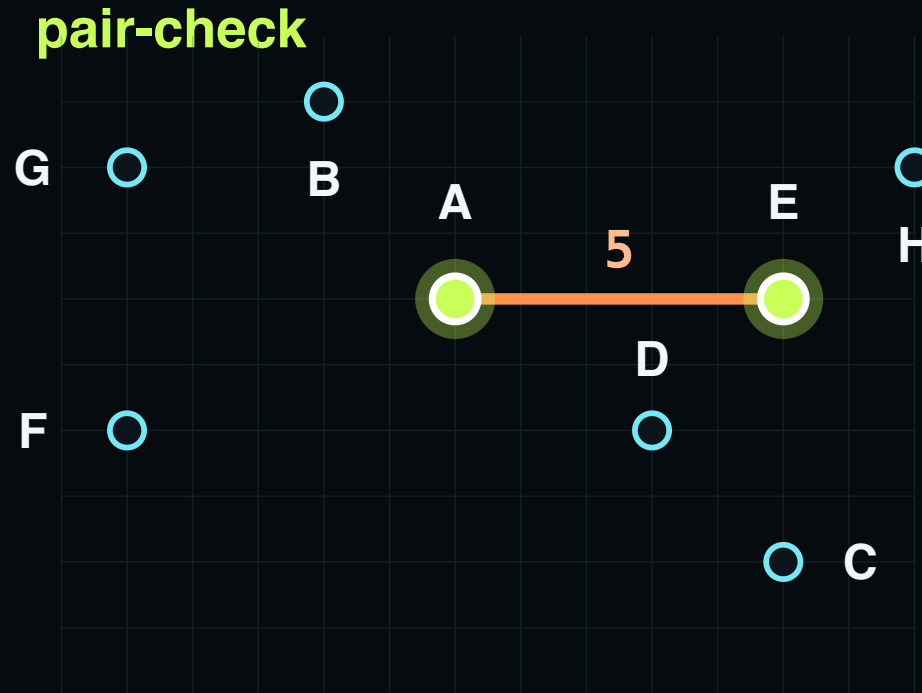
- one distance variable per selected pair
- quadratic count of y variables
- quadratic number of propagators
- (potentially) quadratic domain sizes

DEDICATED CONSTRAINT

$$\text{min_distance}(D, x, z)$$

- single constraint
- global view of problem
- smarter propagation choices

PAIR-CHECK CHECKS ON ASSIGNMENT



z bounds
same scale

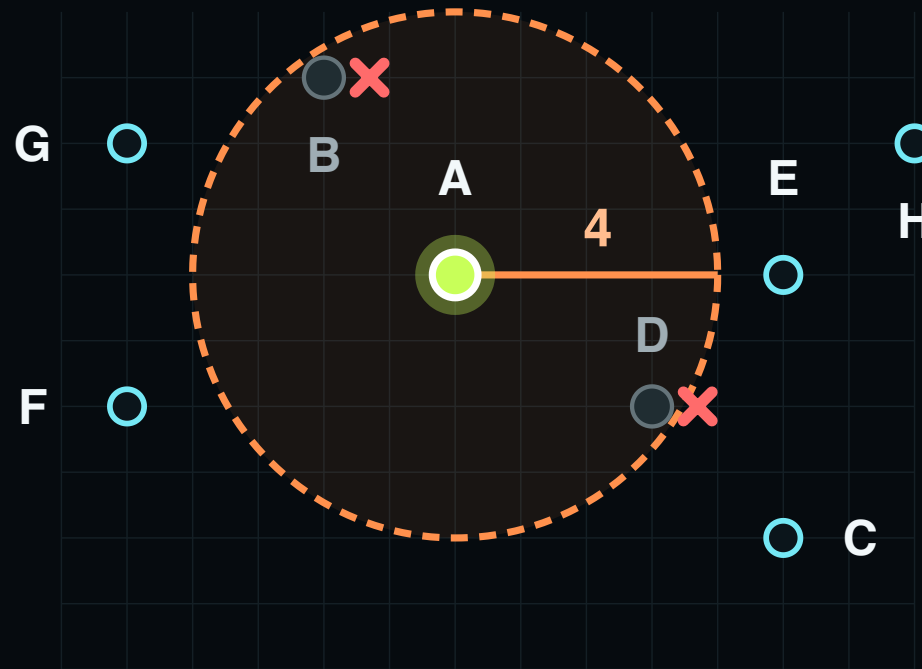
--- previous $z.\max = 7$

$z.\max \leftarrow 5$
 $z.\min = 4$

filled = selected · hollow = possible

A-E is 5, so $z \leq 5$. Another selected pair may still lower z .

FORWARD-BOUND SCANS ON ASSIGNMENT



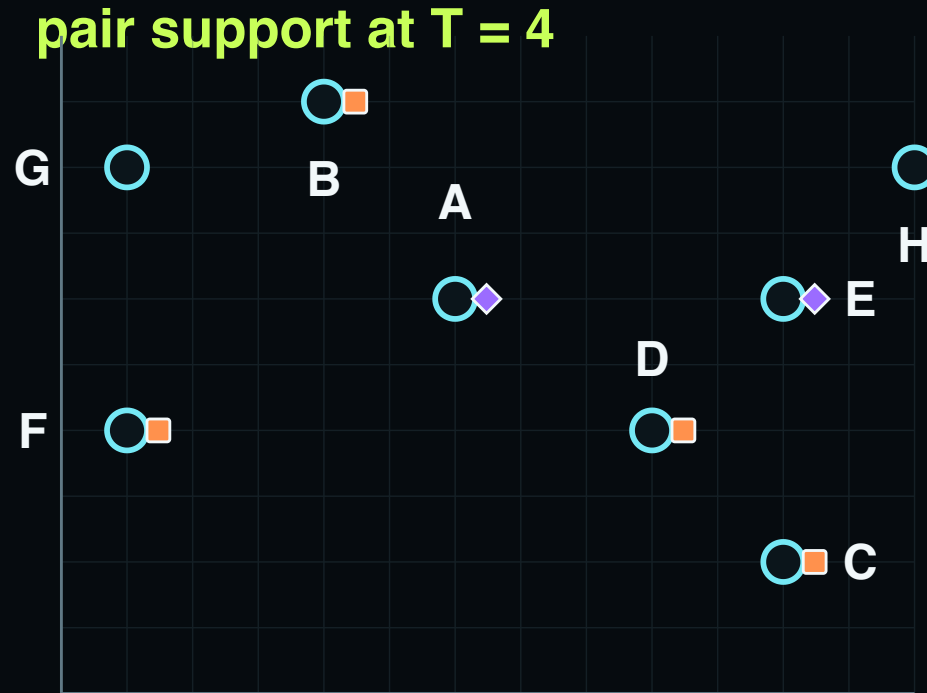
z bounds
same scale

$z.\max = 5$
 $z.\min = 4$

filled = selected · hollow = possible · x = removed

At $T = 4$, B and D are strictly too close to A. Equality stays feasible.

PAIR SUPPORT STARTS ON THE SITE GRAPH



two variable domains

◆ variable x_i

$\text{dom}(x_i)$
 $= \{A, E\}$

■ variable x_j

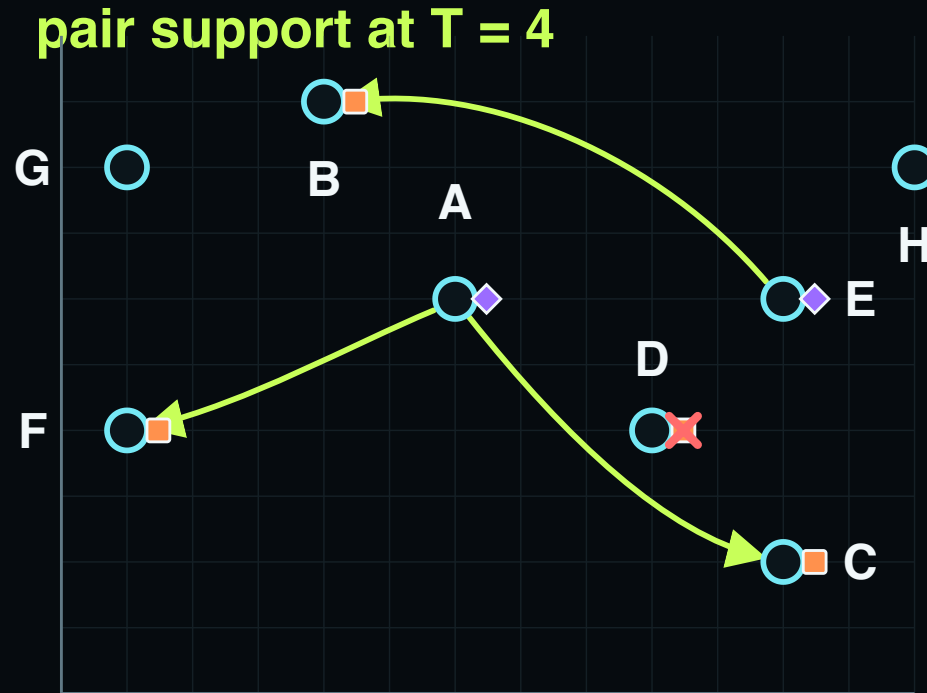
$\text{dom}(x_j)$
 $= \{B, C, D, F\}$

support requires
distance ≥ 4

violet diamond = value of x_i · orange square = value of x_j · no sites selected

Can every value of x_j find a partner at distance at least 4?

PAIR SUPPORT REMOVES D



supported values

◆ variable x_i

$\text{dom}(x_i)$
 $= \{A, E\}$

■ variable x_j

$\text{dom}(x_j)$
 $= \{B, C, \text{D}, F\}$

D has no support

remove $x_j = D$

lime = support · red x = remove $x_j = D$ · site D remains

Thin curves witness B, C, and F; remove the unsupported $x_j = D$.

DETAIL: ONE RELATION, FOUR KERNELS

KERNEL	WHEN IT CAN ACT	WHAT IT DOES
Pair-check	both endpoints fixed	measures the selected pair
Forward-bound	one endpoint fixed	prunes inside the active radius
Full pair support	before either is fixed	finds a partner for every value
Advisors	after relevant domain events	revisits affected pairs and witnesses

Pairwise and global variants package these kernels differently. Matching is a separate upper-bound pass.

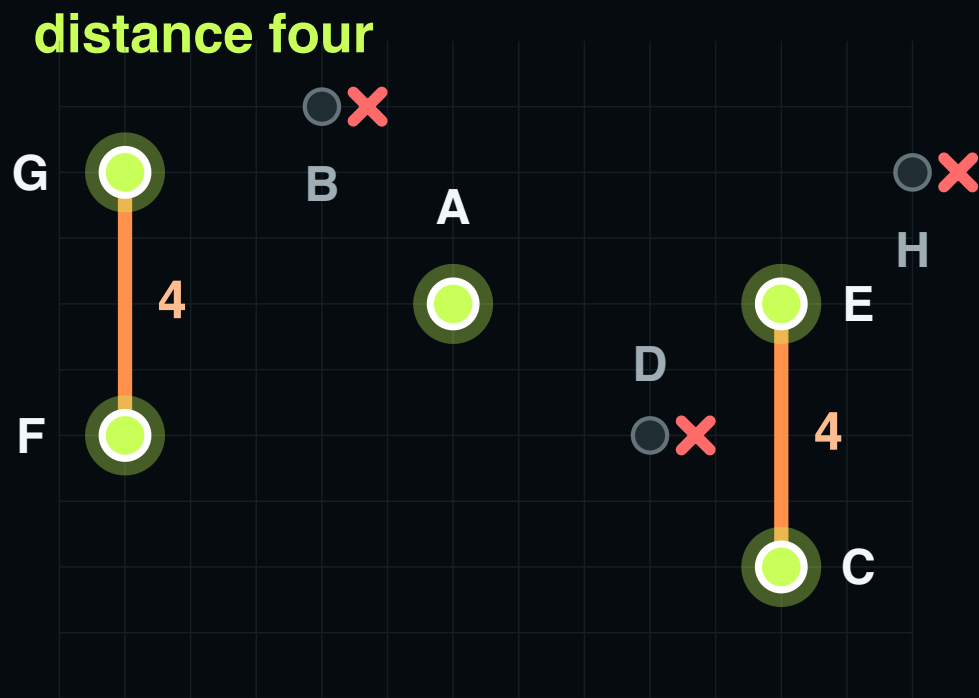
DETAIL: ONE RELATION, THREE MODEL VIEWS



MODEL VIEW	DECISION VARIABLES	WHAT THE PROPAGATOR SEES
Indexed	p site-valued variables	one current site domain per position
Site-boolean	n selection booleans	selected and possible sites
Assignment-boolean	$p \times n$ assignment booleans	a Boolean encoding of each position domain

The pictures use the indexed view. The matching graph uses the union of its current position domains.

DISTANCE FOUR SOLUTION



z bounds
same scale

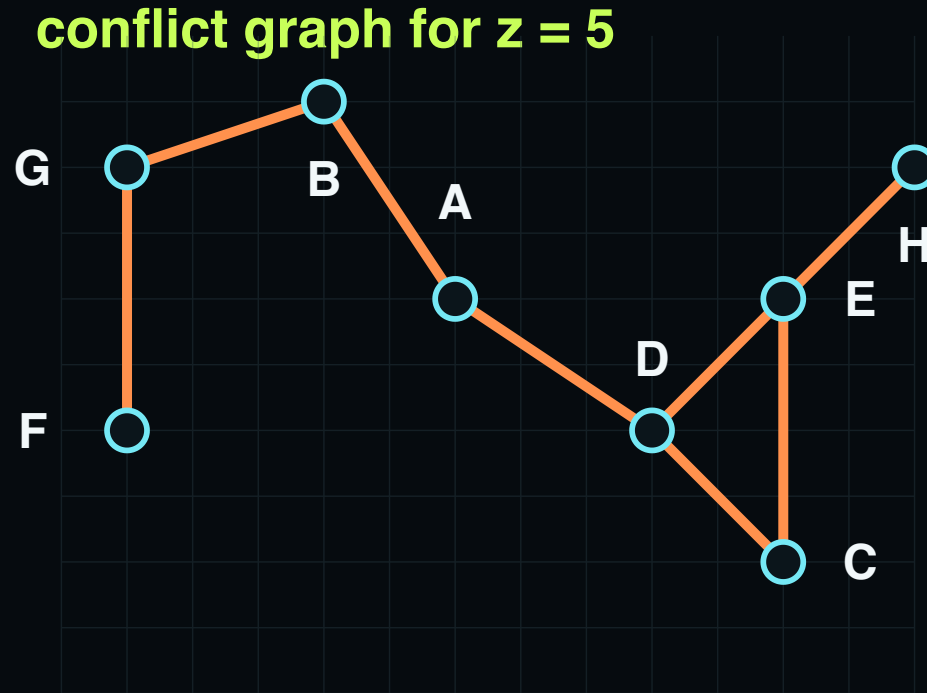
 z.max = 5
z.min = 4

attains 4
can z reach 5?

filled = selected · muted × = no longer possible

Sites A,C,E,F,G reach distance 4. The next distance level is 5.

BUILD THE CONFLICTS FOR $z = 5$



edges that interfere

distance < 5

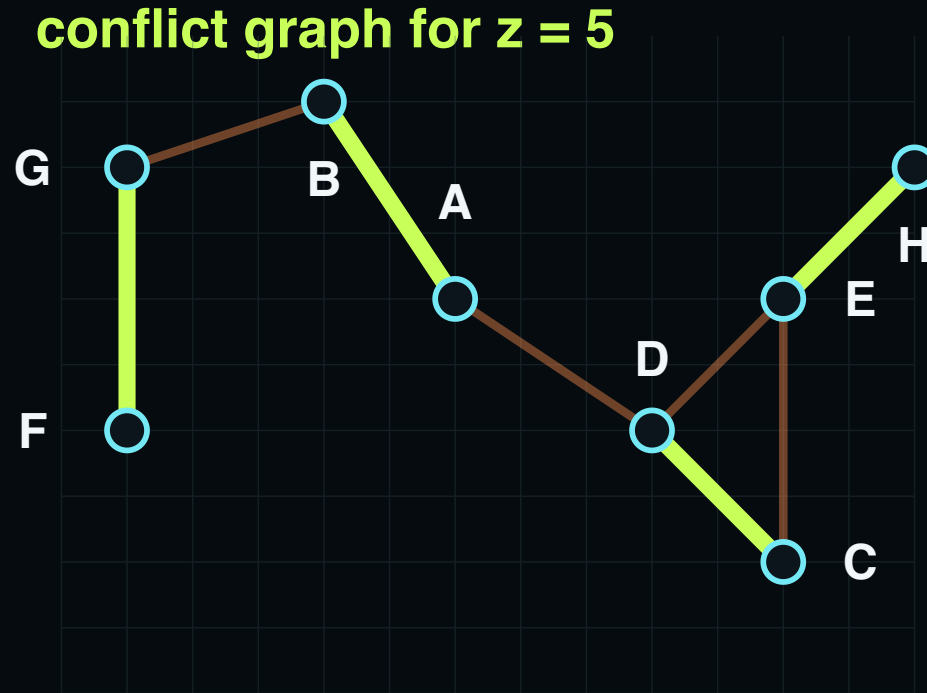
both endpoints cannot
be selected together

in a $z = 5$ solution

orange = cannot coexist if $z = 5$ · no matching marked yet

Join two sites when selecting both would interfere with $z = 5$.

A MATCHING PROVES $z < 5$



four disjoint conflicts

$$|M| = 4$$

omit at least one site
from each matched edge

$$8 - 4 = 4 < p = 5$$

therefore $z < 5$

the next level is 4

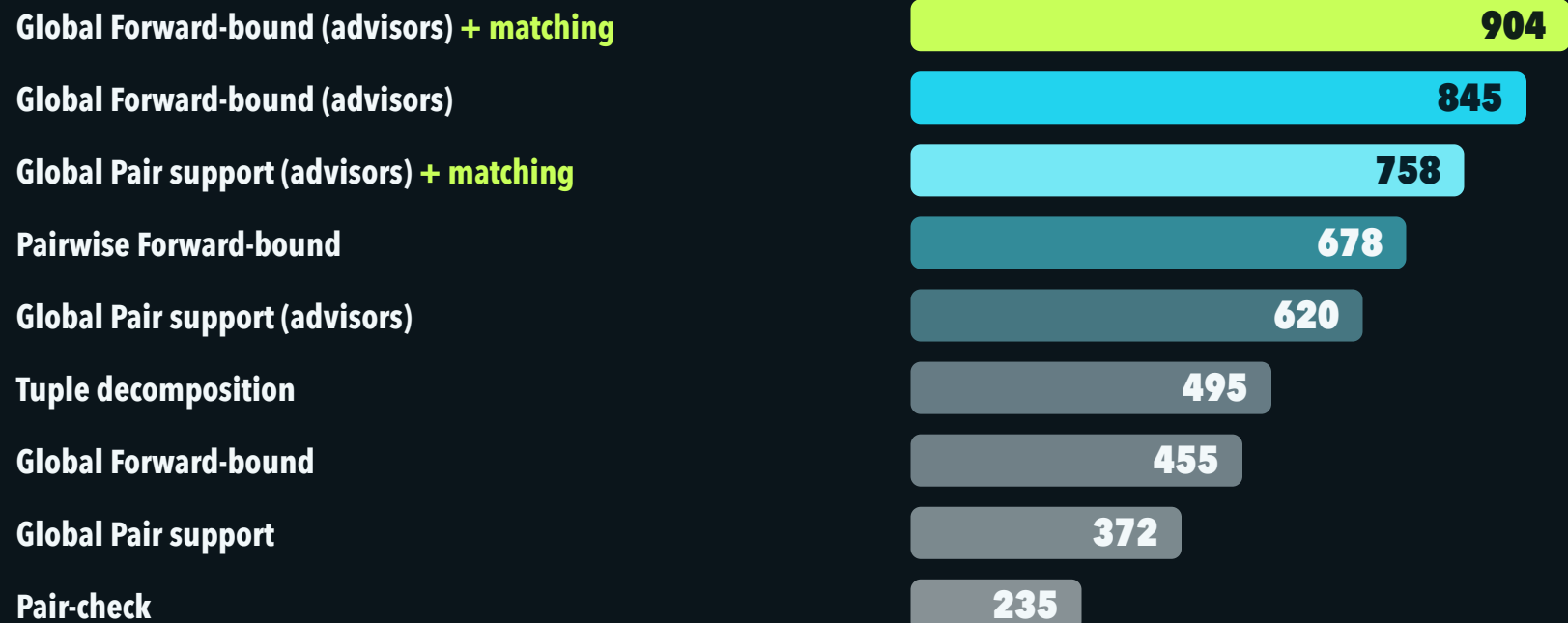
orange = conflict · lime = four-edge matching · proof that $z < 5$

Four matched conflicts leave at most four sites, proving $z < 5$.

ADVISORS AND MATCHING ARE USEFUL

Anytime area score

higher is better



Aggressive heuristic (80) · Safe heuristic (52)



**Some propagators
are easy
to implement**

**Advisors
are useful**

**Avoid quadratic
numbers of
propagators
and variables**

**Matching
gives a useful
upper bound
for *min_distance***